

1 Causes of variation in vaccine efficacy in a wild rodent
2 population

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13 **Individuals vary broadly in their response to vaccination and subsequent chal-**
14 **lenge infection, with poor vaccine responders causing persistence of both**
15 **infection and transmission in populations.**

16 **Keywords:** Helminth; vaccine; immunoparasitology; machine learning; pathway ana-
17 lysis; ageing.

INTRODUCTION

Individuals vary widely in their responses to vaccination and little is known about the causes underlying low vaccine efficacy, nor how effective a vaccine will be “in the real world” against the pathogen for which it has been developed.¹

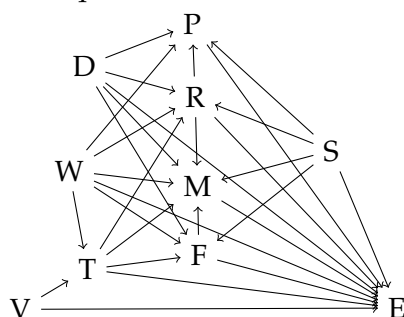
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RESULTS

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METHODS

ACKNOWLEDGEMENTS

COMPETING INTERESTS

48 The authors declare that there are no competing interests.

AUTHOR CONTRIBUTIONS

REFERENCES

- 49 1. Belkin, M. & Niyogi, P. *Using manifold structure for partially labeled classification in Advances in neural*
 50 *information processing systems* (2002), 929–936.

FIGURE LEGENDS

SUPPLEMENTARY INFORMATION

51 **Supplementary Data 1**

52 **Supplementary Data 2**

53 **Supplementary Data 3**

54 **Supplementary Figures**